

A six-vector counterexample to the proposed ℓ_p plank theorem

A negative answer to Question 6.13 of arXiv:1209.1462

Autonomous proof packet — agent_lane_12

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Abstract

Question 6.13 asks for a simultaneous lower-bound vector in the dual of complex ℓ_p whenever the prescribed lower bounds have unit $\ell_{\min\{2,q\}}$ norm. We give a counterexample at the endpoint $p = 1, q = \infty$. Two coordinate functionals and four diagonal functionals in ℓ_1^2 , all with threshold $2/5$, already exclude every point of the unit ball of ℓ_∞ . Their squared thresholds total only $24/25$. A strictly positive geometric tail uses the remaining $1/25$, producing exactly the infinite sequences required in the question. The proof is an elementary four-point covering of the complex projective line, with strict rational margins.

1 The source question

The source is S. Shkarin, *Non-sequential weak supercyclicity and hypercyclicity*, arXiv:1209.1462 [1]. Question 6.13 on PDF page 33 asks whether, for conjugate $p, q, r = \min\{2, q\}$, unit vectors $x_n \in \ell_p$, and positive s_n with $\sum s_n^r = 1$, there must be a unit $y \in \ell_q$ satisfying $|\langle x_n, y \rangle| \geq s_n$ for every n .

----- (n) -----
Proof. The case $1 \leq p \leq 2$ follows from Proposition 6.11. If $p > 2$, we can take $x_n = c_n e_n$ and apply Lemma 2.2. $\square$

Corollary 6.12 for $p = 2$ and $p = \infty$ implies that conditions on the growth of $\|x_n\|$ in Propositions 5.2 and 5.4 are best possible. Proposition 5.3 and Corollary 6.12 lead to the natural conjecture that the best possible condition on the growth of $\|x_n\|$ implying weak closeness of $\{x_n : n \in \mathbb{N}_0\}$ in ℓ_p is (38) with $a = \min\{2, q\}$. In order to prove this conjecture it would suffice to answer the following question affirmatively.

QUESTION 6.13. Let $1 \leq p < \infty$, $\{x_n\}_{n \in \mathbb{N}_0}$ be a sequence in the unit sphere of a the complex Banach space ℓ_p and $\{s_n\}_{n \in \mathbb{N}_0}$ be a sequence of positive numbers such that $\sum_{n=0}^{\infty} s_n^r = 1$, where $r = \min\{2, q\}$ and $\frac{1}{p} + \frac{1}{q} = 1$. Does there exist $y \in \ell_q$ such that $\|y\|_q = 1$ and $|\langle x_n, y \rangle| \geq s_n$ for each $n \in \mathbb{N}_0$?

Note that an affirmative answer to this question would also provide an interesting generalization of Ball's theorem (Theorem B1) and possibly lead to further applications in convex analysis.

6.4 Sequential weak topology

We use the bilinear dual pairing $\langle x, y \rangle = \sum_j x(j)y(j)$. The same construction works for the conjugate convention because the set of fourth roots of unity is closed under conjugation.

2 Counterexample

Theorem 1. *The answer to Question 6.13 is negative. In fact, for $p = 1$, $q = \infty$ there are sequences $(x_n)_{n \geq 0}$ in the unit sphere of complex ℓ_1 and positive numbers $(s_n)_{n \geq 0}$ such that*

$$\sum_{n=0}^{\infty} s_n^2 = 1,$$

but no $y \in \ell_{\infty}$ of norm one satisfies $|\langle x_n, y \rangle| \geq s_n$ for every n .

Proof. Let e_1, e_2 be the first two coordinate vectors and put

$$\Omega = \{1, -1, i, -i\}.$$

Take as the first six vectors

$$x_0 = e_1, \quad x_1 = e_2, \quad \{x_2, x_3, x_4, x_5\} = \left\{ \frac{e_1 - \bar{\omega}e_2}{2} : \omega \in \Omega \right\}. \quad (1)$$

Every vector in (1) has ℓ_1 norm one. Set

$$s_0 = \cdots = s_5 = \frac{2}{5}. \quad (2)$$

We first show that the six inequalities from (1)–(2) already have no simultaneous solution in the closed unit ball of ℓ_{∞} . Let $\|y\|_{\infty} \leq 1$. If either $|y(1)| < 2/5$ or $|y(2)| < 2/5$, one of the two coordinate inequalities fails. We may therefore assume both coordinate moduli are at least $2/5$.

Let $m = \max\{|y(1)|, |y(2)|\} \leq 1$. Divide the first two coordinates by a common unimodular scalar. Using the larger coordinate as the projective denominator, their ratio has the form $z = \rho e^{i\theta}$ with

$$\frac{2}{5} \leq \rho \leq 1. \quad (3)$$

Choose $\omega \in \Omega$ whose argument differs from θ by at most $\pi/4$. Whether the first or second coordinate is the denominator, one of the four diagonal correlations has modulus

$$\frac{m}{2} |z - \omega|.$$

Since $\cos(\theta - \arg \omega) \geq 1/\sqrt{2}$,

$$|z - \omega|^2 \leq 1 + \rho^2 - \sqrt{2}\rho. \quad (4)$$

The right side of (4) is convex in ρ , so its maximum on the interval in (3) occurs at an endpoint. The two endpoint values satisfy

$$2 - \sqrt{2} < \frac{16}{25}, \quad \frac{29}{25} - \frac{2\sqrt{2}}{5} < \frac{16}{25}. \quad (5)$$

Indeed, the inequalities in (5) reduce respectively to $2 > (34/25)^2$ and $2 > (13/10)^2$. It follows that $|z - \omega| < 4/5$, and therefore the selected diagonal correlation is strictly smaller than

$$\frac{m}{2} \frac{4}{5} \leq \frac{2}{5}.$$

Thus at least one of the first six required inequalities always fails.

It remains only to make the sequence of thresholds infinite, strictly positive, and exactly normalized. For $k \geq 1$, define

$$x_{5+k} = e_1, \quad s_{5+k} = \frac{\sqrt{3}}{5} 2^{-k}. \quad (6)$$

All the x_n are unit vectors of ℓ_1 and all s_n are positive. Moreover,

$$\sum_{n=0}^{\infty} s_n^2 = 6 \left(\frac{2}{5} \right)^2 + \frac{3}{25} \sum_{k=1}^{\infty} 4^{-k} = \frac{24}{25} + \frac{1}{25} = 1.$$

The tail cannot restore a simultaneous solution already excluded by the first six constraints. Zero-padding the vectors embeds the construction from $\mathbb{C}^2$ into the source's infinite-dimensional sequence spaces. $\square$

3 Geometry, checks, and scope

The two coordinate functionals say that a candidate must lie in the projective annulus where the smaller of its first two coordinates is not too small. The other four functionals have zero directions at the fourth roots of unity. Every point of that annulus is within Euclidean distance $4/5$ of one of those roots. The factor $1/2$ in the ℓ_1 normalization then makes the corresponding correlation smaller than $2/5$. The endpoint checks in (5) are strict, so equality cases and nonattainment of the ℓ_∞ norm cause no difficulty.

The companion script `code/verify_counterexample.py` checks the two exact rational margins, the norm and budget identities, and a dense diagnostic grid for (4). The proof itself is exact and does not depend on the numerical grid.

This gives a full negative answer to the universal statement of Question 6.13. It does *not* disprove the separate weak-closedness conjecture that motivated the question: the source says only that an affirmative answer to Question 6.13 would suffice for that conjecture.

Bounded novelty checks on 11 August 2026 covered the run's cheap indexes and arXiv/web searches by exact title, arXiv id, question number, and the ℓ_p plank formulation. They found the source and later plank surveys, but no cited resolution of Question 6.13 or this six-vector example. This is a search report, not a priority claim.

Human-review recommendation. Accept as a candidate full counterexample. The only substantive checks are the four-root covering estimate (4)–(5) and the positive-tail normalization; both are elementary and strict.

References

- [1] S. Shkarin, *Non-sequential weak supercyclicity and hypercyclicity*, arXiv:1209.1462 (2012), Question 6.13.